

Searching for Clues in the Question

You really thought I was gonna leave my name on huh

August 30, 2019

Contents

1	Introduction	1
2	There is no such thing as a coincidence	1
3	That's definitely not the extremal case	5
4	Isn't that a strange way to phrase it?	8

1 Introduction

This script is written with the objective of helping the reader to identify recurring features in combinatorics and number theory questions that can be exploited to understand how to solve the question. There is no theory to be learned here; the key idea the reader should take away is to never accept things for granted in the statement of an olympiad problem. Questioners love to seek simplicity in the problem statements, as after all, this is where the great beauty of mathematical olympiads lies: in giving clear and comprehensible problems with solutions that are elegant and that any high-schooler could understand, but which are difficult to resolve. If you see a problem statement that seems more complicated than it needs to be, then that is a sign that things are not as they seem.

This is by no means a comprehensive guide on the matter, nor does it claim to help you crush the problem every time. Often, it will only give you a little insight - a little guide as to what your approach should be - but sometimes, with the right ideas, it can utterly crush a high-level problem, like with the last example in this script.

2 There is no such thing as a coincidence

Every contestant has had a moment where they are reading a usually when the number being used is significant for some reason. There is no such thing as a coincidence. If the problem is using the number 1156, then the fact that $34^2 = 1156$ is clearly instrumental to solving it. Nobody just *happens to stumble* upon the value 1156. Let's kick things off with a simple example.

Example 1. *Prove that the number 1280000401 is composite.*

1280000401 is a strange number. At first glance, it has no apparent significance: its only semi-remarkable feature is that it seems to contain a lot of zeroes. However, the key to this problem lies not in looking at the zeroes, but rather the other numbers. We see that there is a 128 at the start, and a 4 and a 1 at the end. By now, you should be familiar enough with the integers to recognize instantly that $128 = 2^7$. (If this does not occur to you, then do not fret; it will come with practice and experience. Nevertheless, it may be useful to know the first few powers of 2 and 5, since these are important components of our base-10 number system.) 4 is also a power of 2 - what a coincidence! - but remember: there are *no* coincidences. Let's try and explore what the significance of this is.

$$\begin{aligned} 1280000401 &= 1280000000 + 400 + 1 \\ &= 2^7 \cdot 10000000 + 2^2 \cdot 100 + 1 \\ &= 2^7 \cdot 10^7 + 2^2 \cdot 10^2 + 1 \\ &= 20^7 + 20^2 + 1 \end{aligned}$$

Amazing: we have managed to transform our bizarre number into a fairly simple expression. The logical next step is to try and factorise the polynomial $P(x) = x^7 + x^2 + 1$, as if we can express this as the product of two polynomials of lesser degree, then by substituting 20 into the said polynomials, we will find factors of 1280000401. This is the hardest part of the problem, but we can use small tricks to help us - for example, $P(x)$ has no factors of the form $(x - k)$, as this would imply $P(k) = 0$, which isn't true for any integer k . This means we can test quadratic polynomials: $x^2 + 1$ does not divide it, there is no point testing $x^2 + x$ as this is the product of $x + 1$ and x which we already know do not divide $P(x)$. Testing $x^2 + x + 1$, we obtain

$$x^7 + x^2 + 1 = (x^2 + x + 1)(x^5 - x^4 + x^2 - x + 1)$$

Plugging 20 into $x^2 + x + 1$, we get 421, which divides 1280000401.

Solution. As $1280000401 = 20^7 + 20^2 + 1$, then consider $P(x) = x^7 + x^2 + 1$. We can factorise $P(x) = (x^2 + x + 1)(x^5 - x^4 + x^2 - x + 1)$, meaning that $(20^2 + 20 + 1)(20^5 - 20^4 + 20^2 - 20 + 1) = 421 \cdot 3040381 = 1280000401$ $\square$

Now for an example from combinatorics.

Example 2 (First Round, 2004). *We have a standard chessboard, on which 17 rooks are placed. Prove that we can choose three of them such that none of them attacks either of the others*

You should immediately be thinking 17 - that's $16 + 1$. This makes using the Pigeonhole Principle irresistible, as a chessboard has 8 rows and 8 columns, and 16 is $8 \cdot 2$. So what do we know, thanks to there being 17 rooks? Well, we know there is a row with at least 3 rooks, and a column with at least 3 rooks. But neither of these actually helps us - that just gives us 3 rooks that are attacking each other.

Stating that we know there are 3 rooks in a row comes from taking eight groups (the eight rows) and stating we know one of these groups must contain 3 rooks. What we want to do, ideally, is to create 8 groups of squares such that a rook stationed on one of the squares attacks none of the others. Thankfully, this is actually quite easy to do; an example is given below, although there are thousands of ways of doing so - the colouration is similar to a sudoku, in that every row and column must contain each colour (or number, as below) exactly once. A simple way to do it is just to notice that no two rooks on a common diagonal will ever be able to attack each other, so we simply need to partition our chessboard into 8 groups of squares using the diagonals.

1	2	3	4	5	6	7	8
2	3	4	5	6	7	8	1
3	4	5	6	7	8	1	2
4	5	6	7	8	1	2	3
5	6	7	8	1	2	3	4
6	7	8	1	2	3	4	5
7	8	1	2	3	4	5	6
8	1	2	3	4	5	6	7

Figure 1: A colouration for Example 2

Solution. Consider the colouration given in Figure 1. By the Pigeonhole principle, 3 of our rooks must all lie on three squares of the same value. As no rook on a square of a given value attacks any other squares of that value, it follows that these three rooks are mutually unattacked. $\square$

In this question, 17 rooks was the extremal case - if we had 16 rooks, the problem no longer holds (by filling 2 columns, for example) whereas for more than 17 the problem holds, but we have more than enough to work with.

However, in combinatorics, the examiners are smarter than this, and will often choose values that do not necessarily represent the extremal cases. If you're feeling confused by what this means, then the best way to understand is through another example.

Example 3 (First Round, 2011). *We write eleven arbitrary integers on a board. Show that we can choose some of them and place + and - symbols between them such that the result is divisible by 2011.*

Your initial thought upon seeing this problem should be: why 2011 and 11? Clearly, the integers are arbitrary, and the more of them we have, the easier it would be to find a group of them resulting in a total divisible by 2011, so we can assume 11 is the *minimum* number of integers we can have that allows for the condition to be fulfilled - if we could do it with 10, then why would they allow us the extra breathing space?

Thinking about what the question means, we see that we are looking for a combination of the integers on the board which yields something that is 0 (mod 2011). Clearly, the more integers we had on the board, the more ways we would have to combine them, and the more chance there would be that we could find something that is indeed divisible by 2011. It doesn't seem like this is a property specific to 2011; rather, it seems like the more numbers we have on the board, the bigger the modulus we could find a "0" for.

So how does 2011 relate to 11? Well, we could say that $2011 = 2000 + 11$, but somehow this doesn't seem to be the general rule. (With 12 numbers on our board, could we only go up to finding something divisible by 2012? With 2, we certainly can't find something that is 0 modulo 2002.) Indeed, after a bit of searching, there appears to be no apparent link. Maybe 2011 isn't the extremal case - maybe we could go even bigger.

Assuming this, we now look for numbers related to 11 that are close to 2011, but not necessarily equal. $11^2 = 121$ is nowhere close; $11^3 = 1331$ is much better but still far off. $11!$ is far too big; it's roughly equal to 40 million. 2^{11} , on the other hand, is 2048 - relatively very close to 2011. What if we try working with that instead?

The new question now is how to come across the value 2^{11} . If we consider all possible ways of summing our numbers, we see that each of the 11 numbers is either added, subtracted, or ignored, which gives 3^{11} summations (although the output values are not necessarily distinct). Either way, we can't really use the Pigeonhole Principle on these 3^{11} values, since it's not at all clear how many times 2011 goes into the total number of different values.

How do we make 2^{11} appear? Ideally, this would happen if we could only do two things to each integer instead of three. Let's try this: consider all the possible values we get when we either *add* or *ignore* each of the 11 integers; this gives us $2^{11} = 2048$ different ways to put them together. By Dirichlet, two of the summations must have the same value modulo 2011, and so their difference is divisible by 2011. Note that this idea is actually the key to the proof, so if it doesn't seem that evident a line of thought, then fear not.

Let's subtract one of our summations from the other, so that the result is divisible by 2011. However, we notice that the result can be composed using the integers on the board! If an integer on the board appears in the first summation but not the second, we add it; if it appears in the second but not the first, we subtract it. If it appears in both or neither, we leave it out. The resultant value is equal to our first sum minus the second one.

Solution. With 11 integers, we have 2^{11} ways to sum them by either adding or ignoring each integer. By the Pigeonhole Principle, two of the values are the same modulo 2011; their difference can be written as a sum of the integers on the board (when either adding, subtracting, or leaving out each integer). $\square$

Here, the proof applies up for divisibility up to 2047; 2011 is not actually the maximal case. However, rest assured that usually the extremal case is usually very close to the one specified in the question. Eagle-eyed readers may notice that 2011 was chosen because this was the year the problem was asked; this is a recurring theme at the Olympiads where the questioners like to shoehorn the year they set the problems into the questions themselves. This is an excellent reason for which you should check the prime factorisation of the year you are in before taking an exam, and also indicates that if the number being used in your problem is the year you are currently in, it may not actually be the extremal case for which the question could be asked - which is the topic of our next section.

3 That's definitely not the extremal case

At some point, you will solve a problem and think "Wait. They asked me to prove that so-and-so was true for n , but in reality they could have taken

a much bigger or a much smaller n , and the solution would have been exactly the same!” This occurs when a problem has a general solution but the examiners decide to insert a specific value into it, often the year in which the problem was posed. By noticing that the statement you have been asked to prove is likely not only true for the given number, you can make it much easier to understand what is going on by taking a much smaller case and solving that first. An excellent example of this follows now.

Example 4. *Does there exist a Fibonacci number that ends in at least 2014 zeroes?*

No prizes for guessing which year this problem was published. Immediately, we can start thinking about the fact that 2014 is quite an arbitrary number, with no apparent significance. It seems quite likely if a Fibonacci number exists that ends in 2013 zeroes, then one also exists ending in 2014 of them - after all, why would *two thousand and thirteen* just happen to be the upper bound? Intuitively, it shouldn't be.

Let's try and tackle this problem for a small number of zeroes - say, two of them - and see if the solution for the small case perhaps gives us a hint on how to handle the bigger problem. Spoiler alert: it helps quite a bit.

Clearly, what we care about here is what the number is modulo 100; we can just consider the Fibonacci sequence (mod 100), and if we find a 0 (apart from the one at the very start) then we are done; this number ends in two zeroes. Let's look at the first few values. Note that often the first term of the Fibonacci sequence is considered to be 1; however, here, we side with the minority and take it as being 0. This will prove to be useful later on.

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 44, 33, 77, 10, 87, 97, 84, 81, 65, 46, 11, ...

There don't appear to be any apparent patterns. Each subsequent number in the sequence is not really computationally intensive; you simply add the two previous values, and take away 100 if the result is greater than 100.

Interestingly, this means for two given consecutive values are always followed by the same value: if we ever have “34, 55” in our sequence then the following value is always going to be 89. This doesn't seem to apply backwards, though - 38 could be preceded by “1, 37” or “2, 36” or “69, 69”; there are a hundred different pairs (literally) that could precede it.

Why restrict ourselves to one number in the sequence, though? Seeing as two consecutive values uniquely determine what comes next, however, they should also uniquely determine what comes *before*. This is clear to see - “34, 55” will always be preceded by 21, as no other integer x between 0 and 99 inclusive satisfies the equation $x + 34 = 55 \pmod{100}$.

Clearly, if two numbers uniquely determine the next number, then they also determine the number after that: “34, 55” will always be followed by “89, 44” - and this also works backwards - it will always be preceded by “13, 21”. This unique determination in both directions is quite a powerful property, and it leads us to want to pair up the numbers, giving the following sequence:

$$(0, 1), (1, 2), (3, 5), (8, 13), (21, 34), (55, 89), (44, 33), (77, 10), (87, 97), \dots$$

Suppose a pair in the sequence occurs twice. This means, firstly, that $(0,1)$ must occur twice as well. (Say the pair that appears twice appears as the n th and $(n+k)$ th terms. As by going back $n-1$ terms from the first time it appears we get to the first term, $(0,1)$, then due to the sequence being uniquely determined in both directions, we will also get to $(0,1)$ by going back $n-1$ terms from the second time it occurs - in other words, $(0,1)$ appears as the k th term.) This implies that the sequence must then be periodic with period k , as going forward $k-1$ terms from a $(0,1)$ takes you to another $(0,1)$, and everything in between is uniquely determined. Periodicity, of course, implies that eventually 0 will appear again (since it appears at the very start). Let us therefore assume that the sequence is not periodic - so that no term occurs twice.

But this is impossible! There are only finitely many possible pairs, as we have a choice of 100 numbers (we are working modulo 100) for each term, with a total of $100 \cdot 100 = 10000$ possibilities is quite a lot, but not quite infinitely many¹. Thus, eventually a pair must repeat, and we have proved that for the small case, the assertion is true: there exists a Fibonacci number ending in 2 zeroes²; indeed, there are infinitely many of them.

Here, the reader may want to use a tissue to wipe their tears at how beautiful this is: this proof works for *any* number of zeroes. All we have to do is consider the sequence modulo 10 to an appropriate power; the sequence is still uniquely determined in both directions and therefore the repetition of a term still implies periodicity, but we will always have a finite number of possible terms. Keep in mind the concept of *unique determination* - it is extremely useful when handling problems on sequences, as it implies either periodicity or that no term occurs twice; it is certainly not a feature unique to the Fibonacci sequence.

Solution. Consider the Fibonacci sequence, taken modulo 10^{2014} . It is clear that given a pair of consecutive terms (a, b) in the sequence, the two following

¹Funnily enough, this applies for all integers, not just 10000.

²The first such number is 9969216677189303386214405760200, the 151st Fibonacci number.

terms are uniquely determined $((a + b, a + 2b), (\text{mod } 10^{2014}))$ as well as the two previous terms $((2a - b, b - a), (\text{mod } 10^{2014}))$. This implies, given a pair of consecutive terms, that the sequence is uniquely determined in both directions around that pair. Furthermore, should a given pair of consecutive terms appear twice in the sequence, then the sequence must be periodic. But this is forcibly true, as there are only a finite number of possibilities for a pair of consecutive terms to take, as we are working modulo 10^{2014} . It follows that the sequence is periodic, and that the number 0 must appear again (it appears at the very start of the sequence); the corresponding number in the true Fibonacci sequence is divisible by 10^{2014} and hence contains at least 2014 zeroes at the end. $\square$

Be careful: when you want to select a smaller case to tackle to find a proof you can generalise, you may not be able to select any small number - sometimes for the proof to work, you need the number in question to have a specific property (e.g. it is even, it is 1 (mod 4), etc.). This is true for the first three questions in the problems section.

4 Isn't that a strange way to phrase it?

Finally, we deal with problems that are not remarkable in terms of the numbers being used themselves, but rather the way the *variables* themselves are expressed. The following is a problem 5 from the IMO, and is an excellent showcase of what you can do by searching for clues in the question. At first glance, nothing is strange about it, but we must keep in mind that every detail given is there for a reason. We won't write the proof separately from the thinking process here, as it is far too long.

Example 5 (IMO 2017). *Let $N \geq 2$. A group of $N(N + 1)$ **football** players, no two of which are the same height, stand in a row. Vladimir Petkovic wants to remove $N(N - 1)$ players to leave a new row of $2N$ players in which the following conditions hold:*

- (1) *no one stands between the two tallest players,*
- (2) *no one stands between the third and fourth tallest players*
- $\vdots$
- (N) *no one stands between the two shortest players.*

Show that this is always possible.

Solution. Now, clearly at the end we will have $2N$ players standing in pairs side by side such that each pair contains two players of consecutive height. Given that the initial number of players is $N(N + 1)$, the problem is practically begging us to take N groups of $N + 1$ players and leave 2 players from each group at the end, in order to fulfil the conditions. After all, the fact that the total number of players is so closely related to how many we have left at the end can't be a coincidence! But which groups? And which players from those groups? Why $N(N + 1)$ players, and not N^2 of them?

Let's consider what the problem wants us to do: we have to leave the players in such a way that for all integers k such that $1 \leq k \leq N$, the $(2k - 1)$ th tallest and $(2k)$ th tallest players remain next to each other. Connecting this with our convenient variables, what would be really nice is if we could take N groups of $N + 1$ players, and leave two players from each group next to each other such that these two players were the $(2k - 1)$ th tallest and $(2k)$ th tallest players in the end.

How can we guarantee that there would be nobody between them in height? The simplest way to do so is take our group of $N + 1$ players as consecutive on the list of players when ordered by height; for example, our first group is the tallest $N + 1$ players, then the second group is the next tallest $N + 1$ players and so forth. Thus, if only two players from each group remain at the end, then they are forcibly the $(2k - 1)$ th tallest and $(2k)$ th tallest players, if they come from group k . We want to remove $N - 1$ players from each of the groups so that the two players from a given group are standing next to each other at the end. Here, the contestant must stumble upon the key idea of the proof, which is to use recursion to make his life easier.

Like from the last section, let's try the trick of tackling smaller cases. We observe that for the case $N = 2$, Vladimir can pull it off. He has 6 players, which he can consider as coming from two groups: the taller three and the smaller three. By the Pigeonhole principle, he can look at the first three players in the line; at least two of them must belong to one of the two groups. Amongst the second three players, at least two of them must belong to the other of the two groups. By keeping these 4 players and kicking out the other two, the problem is solved for $N = 2$; the two smaller players are next to each other, as are the two taller players.

Indeed, this process makes a whole lot of sense - this would explain why we have $N(N + 1)$ players, and not N^2 of them - it was done so we could apply the Pigeonhole principle! Generalising this to the case for N , let's look at the first $N + 1$ players from the left; we know that at least 2 of them must belong to the same group (groups being as defined above), let us call it group A . Let us keep these 2 players; to guarantee they are the only ones left from their group, we remove the $N - 1$ other members of their group.

Now, ideally, if we want to use a recursive process then we want to remove 1 player from each of the other groups, so that we are left with the 2 players from A on the very left, and N players from each of the other $N - 1$ groups. But we can achieve this easily - we can suppose that A is the *first* group for which we get two players, when we begin at the very left and add players one by one. This means we can remove the leftmost player from each of the other groups, as this leaves the two players from A next to each other (if two players from the same group B were between them, then this would contradict with the fact that A is the first group for which two of its players appear when starting from the left, as B has two of its players appear before the second player from A shows up) and we have reduced the size of every other group by one.

Now, as mentioned, we are left with the 2 players from A on the very left, and N players from each of the other $N - 1$ groups. But we can just take the $N(N - 1)$ players next to the two players from A and repeat the same process we just did, removing one group entirely with the exception of two players and one player from each of the other groups, so that the two remaining players are on the left, and we then have $N - 1$ groups of $N - 2$ players. This process can be repeated indefinitely until we are left with just the $2N$ players we wanted at the end, arranged in pairs from each group, meaning we have satisfied the condition, pleasing Herr Petkovic. Maybe he can finally lead Switzerland to international success at the Nations League now. $\square$

This is the final example of this script; it is a perfect example of how simply looking at the values and variables giving to us can send us on a logical path that heads towards the solution. The little lightbulb moment - creating a recursive process - actually comes naturally when you try to put two players from the same group next to each other; certainly, you - the reader - are capable with coming up with it yourself. Technically it should be a hard problem, but if you appreciate the unintended help you can glean from how a problem is phrased, then you can greatly simplify your life. There will not always be a subtle idea you can dig up from the phrasing, and you shouldn't depend on it being there, but learning to spot an unexpected hint when it appears could serve you well one day. Always question why the examiner used the numbers (or variables) he did, if it might work for a smaller case, and never forget to check the prime factorisation of the year you are in. For the record, the year won't be prime until 2027 - which is a long way away, though time always goes faster when you're doing math. Godspeed, soldier.

Problems

You will encounter problems like these everywhere, but here are 11 examples to whet your appetite on, ordered by difficulty.

There is no such thing as a coincidence

1. (*Be careful which coincidence you target!*) Consider ten numbers in the open interval $(1, 55)$. Show that three of these numbers constitute the sides of a triangle.
2. (*Fairly straightforward, Swiss TST 2003.*) Let there be integers $a_1, a_2, \dots, a_{101}$ such that $0 < a_1 < a_2 < \dots < a_{101} < 5050$. Show that we can always take four of these numbers a_k, a_l, a_m, a_n such that $5050 \mid (a_k + a_l - a_m - a_n)$.
3. (*Pretty brutal, USAMO 2012*) A circle is divided into 432 congruent arcs by 432 points. The points are coloured such that 108 points are coloured each in red, blue, green and yellow. Prove that one can choose three points of each color in such a way that the four triangles formed by the chosen points of the same color are congruent.
4. (*The most obvious coincidence in history, USAMO 1998*) We have a 98×98 chessboard. One can select any rectangle with sides on the lines of the chessboard and switch the colours (black becomes white, white becomes black). Prove that the minimum number of rectangles required to change the chessboard to one colour is 98.
5. (*Quite challenging but very beautiful, Swiss Final Round 2016*) Consider 2016 different points in the plane. Prove that there are at least 45 different distances between these points.

Is that really the extremal case?

6. (*Nice and simple, Swiss Preliminaries 2008*) Consider 5 positive divisors of 10^{2008} . Show of these there are two whose product is a square.
7. (*An IMO classic, IMO 1987*) Show there is no function $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ with

$$f(f(n)) = n + 1987 \quad ?$$

8. (*Take the correct small case!*, Swiss Final Round 2012) There are 2012 chameleons are seated around a round table. At the start, all the chameleons are red or green. At the end of every minute, every chameleon whose two neighbours had the same colour changes its colour; the rest are unchanged. Show that after 2012 minutes we have two chameleons that have changed colour the same number of times.

9. (*Medium*, Swiss Final Round 2007) Let $M \subset \{1, 2, 3, \dots, 2007\}$ be a set with the following property: for any three elements of M , there are two such that one divides the other. How many elements at most can M contain?

10. (*Very tough*, Swiss TST 2012) An integer m is a *perfect power* if there exist positive integers a and n such that $n > 1$ and $m = a^n$.

- (a) Show there exists a set of 2012 positive integers, such that the sum of the elements of any subset of this set is not a perfect power.
- (b) Show there exists a set of 2012 positive integers, such that the sum of the elements of any subset of this set is a perfect power.

Isn't that a strange way to phrase it?

11. (*Why use $2n$ instead of n ?*, Swiss TST 2014) Let there be $2n$ counters in a line. A move consists of switching two adjacent tokens. How many moves must we make at minimum in order for every token to have been once at the start and the end of the line?